



1 Decay Scheme

^{33}P disintegrates 100% by β^- emission to the ground state of ^{33}S .

Le ^{33}P se désintègre exclusivement par émission β^- vers le niveau fondamental du ^{33}S .

2 Nuclear Data

$$T_{1/2}(^{33}\text{P}) : 25,38 \quad (5) \quad \text{d}$$

$$Q^-(^{33}\text{P}) : 248,5 \quad (11) \quad \text{keV}$$

2.1 β^- Transitions

	Energy (keV)	Probability (%)	Nature	log ft
$\beta^-_{0,0}$	248,5 (11)	100 (0)	Allowed	5,033 (7)

3 Atomic Data

3.1

$$\omega_K : 0,0804 \quad (19)$$

$$\bar{\omega}_L : 0,00093 \quad (18)$$

$$n_{KL} : 1,807 \quad (7)$$

3.1.1 X Radiations

		Energy (keV)	Relative probability
X _K	Kα ₂	2,30667	50,53
	Kα ₁	2,30787	100
	Kβ ₁	2,4641	9,33
X _L	Lℓ	0,1513	
	Lη	0,1531	
	Lβ	0,225 - 0,2272	

3.1.2 Auger Electrons

		Energy (keV)	Relative probability
Auger K			
	KLL	1,985 - 2,119	100
	KLX	2,227 - 2,300	12,4
	KXY	2,440 - 2,456	0,38
Auger L		0,1326 - 0,2132	

4 Electron Emissions

		Energy (keV)	Electrons (per 100 disint.)
$\beta^-_{0,0}$	max:	248,5 (11)	100
	avg:	75,91 (38)	

5 Main Production Modes

- $\left\{ \begin{array}{l} {}^{33}\text{S}(\text{n,p}){}^{33}\text{P} \\ \text{Possible impurities: } {}^{32}\text{P} \text{ (fast neutrons), } {}^{35}\text{S} \text{ (thermal neutrons)} \end{array} \right.$

${}^{36}\text{Cl}(\text{n},\alpha){}^{33}\text{P}$

${}^{30}\text{Si}(\alpha,\text{p}){}^{33}\text{P}$

${}^{34}\text{S}(\text{n,d}){}^{33}\text{P}$

6 References

- P.K.SHELINE, R.B.HOLTZMAN, C.Y.FAN. Phys. Rev. 83 (1951) 919
(Half-life, End-point energy)
- E.N.JENSEN, R.T.NICHOLS, J.CLEMENT, A.POHM. Phys. Rev. 85 (1952) 112
(Half-life, End-point energy)
- T.WESTERMARK. Phys. Rev. 88 (1952) 573
(Half-life, End-point energy)
- R.T.NICHOLS, E.N.JENSEN. Phys. Rev. 94 (1954) 369
(Half-life, End-point energy)
- B.ELBEK, K.O.NIELSEN, O.B.NIELSEN. Phys. Rev. 95 (1954) 96
(End-point energy)
- T.WESTERMARK. Arkiv Fysik 7 (1954) 87
(Half-life, End-point energy)
- J.E.RUSSELL. Bull. Am .Phys. Soc. 3 (1958) 61
(Half-life, End-point energy)
- I.FOGELSTROM-FINEMAN, T.WESTERMARK. Acta Chem. Scan. 14 (1960) 2046
(Half-life)
- P.M.ENDT, C.VAN DER LEUN. Nucl. Phys. A 105 (1967) 1
(Evaluation)
- S.A.REYNOLDS, J.F.EMERY, E.I.WYATT. Nucl. Sci. Eng. 32 (1968) 46
(Half-life)
- F.LAGOUTINE, J.LEGRAND, C.PERROT, J.P.BRETHON, J.MOREL. Int. J. Appl. Radiat. Isotop. 23 (1972) 219
(Half-life)
- P.POLAK, L.LINDNER. Radiochimica Acta 35 (1984) 23
(End-point energy)
- E.SCHÖNFELD, H.JANSSEN. Nucl. Instrum. Methods A369 (1996) 527
(Atomic data)
- M.-M.BÉ, V.CHISTÉ, C.DULIEU, E.BROWNE, V.CHECHEV, N.KUZMENKO, R.HELMEYER, A.NICHOLS, E.SCHONFELD, R.DERSCH. Monographie BIPM-5 Vol. 1 (2004)
(Evaluation)
- M.WANG, W.HUANG, F.G.KONDEV, G.AUDI, S.NAIMI. Chinese Physics C 45 (2021) 030003
(Q-value)
- X.MOUGEOT. Appl. Radiat. Isotopes 201 (2023) 111018
(BetaShape)

